Data Cleaning and Missing Value Handling Report

Identifying Missing Values:

We used isnull().sum() in **pandas** to count missing entries and calculated percentages per column. Visualizations with **seaborn** and **missingno** heatmaps highlighted where missingness was concentrated. Features such as Lot Frontage, Alley, and PoolQC showed high missing rates.

Cleaning Methods Used:

1. **Duplicates:** Removed using df.drop_duplicates().
2. **Categorical Values:** Standardized inconsistent labels (e.g., "yes", "Y" → Yes; "none", "NA" → NaN). Code-like values (TA, Ex) were kept uppercase; descriptive values converted to Title Case.
3. **Numerical Imputation:** Missing values filled with median, chosen for robustness to skew and outliers.
4. Categorical Imputation:
    - Columns where missing meant absence (Alley, Fence, PoolQC) filled with "Unknown".
    - Others filled with **mode** (most frequent value).
5. **Bonus:** Tested KNN imputation for numeric data to capture neighborhood patterns.

Challenges and Insights:

High-missing columns were better handled with "Unknown" rather than dropped, preserving information. Median imputation maintained realistic numeric distributions. Categorical codes required careful handling to avoid distorting domain-specific meanings. Imputation improves model stability by preventing row loss, though it introduces slight estimation bias.

Conclusion

The cleaned dataset is now free of duplicates and missing values, with consistent categories and robust imputations. These steps enhance model reliability, ensure consistent feature interpretation, and provide a stronger foundation for predictive analysis.